Tables and graphs

1

a $y = 1.35x$

b 79.65 m

c The change in depth per minute.

2

a

Number of triangles (t)	1	2	3	5	10	20
Number of matches (m)	3	6	9	15	30	60

b $m = 3t$

c 63

3

a

Number of minutes passed (x)	0	5	10	15	20
Amount of fuel left in tank (y)	80	70	60	50	40

b $y = 80 - 2x$

c $x = 40$

d The amount of fuel is decreasing at a constant rate.

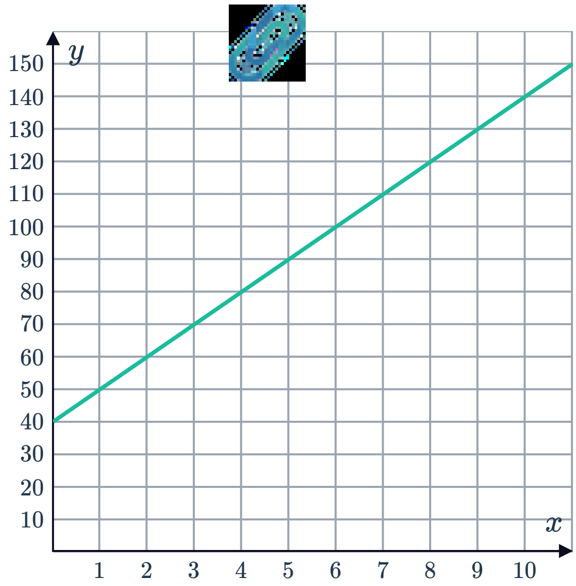
4

a

Number of hours passed (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)	40	50	60	70	80	85	140

b $y = 10x + 40$

c



5

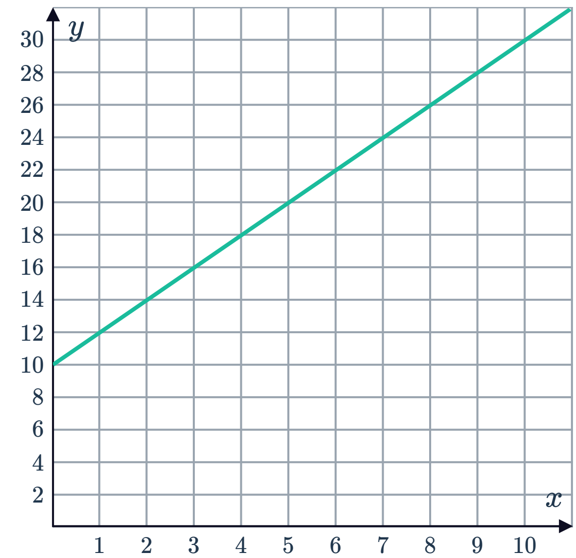
a

Number of hours passed since midnight (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)	10	12	14	16	18	19	30

b $y = 2x + 10$

c 5 hours

d

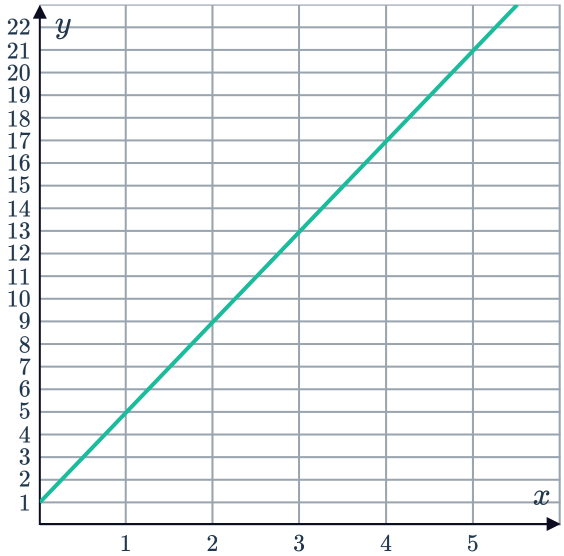


6

a $4^{\circ}\mathrm{C}$

b $y = 1^{\circ}\mathrm{C}$

c $y = 4x + 1$

d


7

a -0.2 seconds

b $y = -0.2x + 9$

c 8.7 seconds

d $x = 2.5$ hours

Word problems

8

i $\dfrac{9}{(x+3)} = \dfrac{6}{(x-3)}$

ii $x = 15$

9 $x = 2500$

10

a $p = 164$

b $\$436$

11 $x = 521$

12 $n = 210$

13 $n = 16$ minutes

14 $t = 6$ hours

15

a $y = 15x + 550$

b $y = \$1090$

16

a $n = 13$

b 26 kph

17

a $n = 50$

b 62 kph

Geometric applications

18 $h = 11 \text{ cm}$

19 $b = 19 \text{ mm}$

20 $x = 8 \text{ cm}$

21 $y = 8 \text{ cm}$

22 $a = 4 \text{ m}$

23 $y = 8$

24 $x = 5$

25 $x = 13$